

LAB 3: ARCHITECTURE SELECTION & JUSTIFICATION

Micro-Lending & Peer Credit Risk Assessor

Student Name: Sakshi Kiran

SRN: PES1UG24AM242

Project

Micro-Lending & Peer Credit Risk Assessor

Architecture Selection

Selected Architecture: Layered Architecture

We chose Layered Architecture for the Micro-Lending & Peer Credit Risk Assessor. The architecture organizes

the system into separate presentation, business, and data responsibilities, providing a clear structure for the

system's major functions.

Justification

Reason 1 - Separation of Responsibilities:

The architecture separates user-facing interactions from business

functions such as credit-risk assessment, loan request and review, repayment processing, and default

monitoring. This makes the responsibilities of the system easier to understand and maintain.

Reason 2 - Suitable for the Project:

The micro-lending system contains closely related functions that work

together through a common application. A layered structure provides an organized design without introducing

the additional operational complexity associated with independently deployed microservices.

Security Advantage:

Sensitive borrower financial information can be handled through controlled business and data layers. This

provides a clear place to apply access controls and helps prevent direct, uncontrolled access to protected

information.

Performance Benefit:

Related credit-risk, loan, and repayment operations can be processed within the application with limited

unnecessary communication between independently deployed services. This supports responsive processing

for risk assessment and repayment-related operations.

Conclusion:

Layered Architecture is selected because it provides a clear separation of responsibilities for the

micro-lending system while supporting controlled handling of financial information and efficient processing of its

core business o

